

Multi-Stage Adaptive Alarm System:

Revised Technical Specifications

1. Overview

This document defines the revised trigger parameters, boundary calculations, and behavioral escalation rules governing the Alarma multi-stage alarm system. The specifications are grounded in real Metro Manila public transportation operating conditions. Key revisions from the previous version include the removal of the fixed 1.0 km distance floor, removal of the 4-minute ETA trigger, removal of time-based stage escalation, and adoption of a fully speed-based boundary system for all three alarm stages.

2. Rolling Average Speed

Rather than using raw instantaneous GPS speed, the system computes a rolling average over the most recent 90 seconds, sampled every 5 seconds:

$$\text{Rolling Average Speed} = \text{Sum of 18 readings} \div 18$$

This produces 18 readings per window (90 sec ÷ 5 sec = 18), dampening stop-and-go noise. A momentary stop at a red light does not collapse the boundary, and a brief burst of speed does not inflate it. All boundary calculations use this rolling average speed.

3. Adaptive Reaction Window

3.1 Default Window

For first-time users (Trips 1–3), a default reaction window of 8 minutes is applied. During these trips the system records the user's response time (how quickly Stage 1 is dismissed) without adjusting the window.

3.2 Adaptive Window Formula

From Trip 4 onward, the window is calculated as:

$$\text{Window (min)} = \text{avg}(\text{last 3 response times} \div 60) + 2\text{-min buffer}$$

The 2-minute buffer accounts for the physical disembarkation process after waking: orienting, gathering belongings, signaling the driver, moving to the door, and waiting for the vehicle to slow down. Response time alone only measures when the user tapped dismiss, not how long it takes to physically get off.

The window is clamped as follows:

- Minimum: 3 minutes — ensures sufficient disembarkation time even for the fastest responders
- Maximum: 8 minutes — no user should need more lead time than a first-time user

Table 1: Example Adaptive Window Calculation

Trip	Response Time	Window Calculation	Window Applied	Note
1	—	Default	8 min	First time
2	10 sec	Not enough data	8 min	Collecting
3	12 sec	Not enough data	8 min	Collecting

Trip	Response Time	Window Calculation	Window Applied	Note
4	8 sec	$\text{avg}(10,12,8) \div 60 + 2 = 2.167 \rightarrow 3 \text{ min}$	3 min	Adaptive

4. Speed-Based Boundary Calculation

4.1 Stage 1 Boundary

Stage 1 fires when the vehicle’s remaining distance to the destination crosses the following boundary, recalculated every 5 seconds:

$$\text{Stage 1 Boundary (km)} = \text{Rolling Avg Speed (km/h)} \times \text{Window (min)} \div 60$$

A ceiling of 5.0 km is enforced to prevent the alarm from firing at an unreasonably early distance on free-flowing roads. No fixed distance floor is applied — slower speeds naturally produce smaller boundaries, and the three-stage escalation system provides sufficient protection for low-speed edge cases.

4.2 Stage 2 and Stage 3 Boundaries

Rather than using fixed time intervals between stages, Stages 2 and 3 use their own speed-based boundaries derived as equal thirds of the full boundary. All three boundaries are continuously recalculated every 5 seconds using the current rolling average speed:

$$\begin{aligned} \text{Stage 2 Boundary (km)} &= \text{Rolling Avg Speed} \times \text{Window} \div 60 \times (2/3) \\ \text{Stage 3 Boundary (km)} &= \text{Rolling Avg Speed} \times \text{Window} \div 60 \times (1/3) \end{aligned}$$

This means if the vehicle accelerates after Stage 1 fires, the Stage 2 and Stage 3 boundaries expand accordingly — causing earlier escalation without any time-based logic. The system checks all boundaries every 5 seconds alongside the rolling average speed update.

Table 2: Boundary Values Across Speed Ranges (3-min Window)

Speed (km/h)	Stage 1 Boundary	Stage 2 Boundary	Stage 3 Boundary	Condition
5 km/h	0.25 km	0.17 km	0.08 km	Heavy traffic
15 km/h	0.75 km	0.50 km	0.25 km	Typical urban
25 km/h	1.25 km	0.83 km	0.42 km	Moderate flow
40 km/h	2.0 km	1.33 km	0.67 km	Clear road
100+ km/h	5.0 km (ceiling)	3.33 km	1.67 km	Ceiling applied

5. Alarm Stage Trigger Specifications

All three stages are triggered purely by speed-based boundary crossings, checked every 5 seconds. There are no fixed time intervals between stages. Each stage fires as soon as the remaining distance drops below its continuously recalculated boundary.

Table 3: Alarm Stage Trigger Specifications

Stage	Name	Trigger Condition	Alert Mode	Dismissal
Stage 1	Gentle Alert	Remaining distance < Speed × Window ÷ 60	Soft vibration + low-volume audio	Tap to dismiss
Stage 2	Louder Alert	Remaining distance < Speed × Window ÷ 60 × (2/3)	Strong vibration + full-volume audio	Tap to dismiss
Stage 3	Emergency Alert	Remaining distance < Speed × Window ÷ 60 × (1/3)	Full-screen takeover, maximum volume, continuous vibration	Slide-to-dismiss required
Overshoot	Overshoot Alert	Consecutive GPS readings confirm increasing distance from destination (+0.5 km past stop)	Full-screen alert + Google Maps return route	Opens Google Maps automatically

6. Worked Example

6.1 Setup

- First-time user → Window = 8 minutes
- Starting speed = 15 km/h
- Vehicle gradually accelerates after Stage 1 fires
- Rolling average updated every 5 seconds (18 readings)

6.2 Stage 1 Trigger

At 15 km/h with 8-minute window:

Stage 1 Boundary = 15 × 8 ÷ 60 = 2.000 km

Stage 1 fires when remaining distance reaches 2.000 km. Commuter does not wake up.

6.3 Stage 2 Trigger (Vehicle Accelerating)

After Stage 1 fires, speed gradually increases. The system recalculates Stage 2 boundary every 5 seconds. Stage 2 fires when remaining distance < Stage 2 boundary:

Table 4: Stage 2 Escalation Tracking

Time	Rolling Avg	Stage 2 Boundary	Remaining Distance	Stage 2 Fires?
0 sec	15.00 km/h	1.333 km	2.000 km	No
5 sec	15.17 km/h	1.348 km	1.979 km	No
10 sec	15.50 km/h	1.378 km	1.956 km	No
15 sec	16.00 km/h	1.422 km	1.929 km	No
20 sec	16.67 km/h	1.481 km	1.897 km	No
25 sec	17.33 km/h	1.541 km	1.862 km	No
30 sec	18.17 km/h	1.615 km	1.820 km	No
35 sec	19.17 km/h	1.704 km	1.771 km	No
40 sec	20.17 km/h	1.793 km	1.722 km	YES — 1.722 < 1.793

Stage 2 fires at 40 seconds when the remaining distance (1.722 km) falls below the recalculated Stage 2 boundary (1.793 km). Note this is driven purely by the expanding boundary, not a timer.

6.4 Stage 3 Trigger

Stage 3 boundary continues to be recalculated every 5 seconds. Stage 3 fires when remaining distance < Stage 3 boundary:

Table 5: Stage 3 Escalation Tracking

Time	Rolling Avg	Stage 3 Boundary	Remaining Distance	Stage 3 Fires?
40 sec	20.17 km/h	0.896 km	1.722 km	No
45 sec	21.00 km/h	0.933 km	1.673 km	No
50 sec	21.83 km/h	0.970 km	1.624 km	No
55 sec	22.50 km/h	1.000 km	1.575 km	No
60 sec	23.17 km/h	1.030 km	1.523 km	No
65 sec	23.83 km/h	1.059 km	1.468 km	No
70 sec	24.33 km/h	1.081 km	1.411 km	No
75 sec	24.83 km/h	1.104 km	1.352 km	No
80 sec	25.17 km/h	1.119 km	1.292 km	No
85 sec	25.50 km/h	1.133 km	1.230 km	No
90 sec	25.67 km/h	1.141 km	1.166 km	No
95 sec	25.83 km/h	1.148 km	1.101 km	YES — 1.101 < 1.148

6.5 Summary

Event	Time Elapsed	Remaining Distance
Stage 1 fires	0 sec	2.000 km
Stage 2 fires	40 sec	1.722 km
Stage 3 fires	95 sec	1.101 km

7. Summary of Revisions from Previous Version

Feature	Previous Version	Revised Version
Distance floor	Fixed 1.0 km	Removed — speed handles naturally
Stage escalation	Time-based (30 sec intervals)	Speed-based boundary (2/3 and 1/3)
ETA trigger	4-minute ETA failsafe	Removed — 3 stages handle this
Speed-up rule	50% threshold shortens to 15 sec	Removed — continuous recalculation handles it
Boundary recalculation	Stage 1 only	All 3 stages, every 5 seconds
Adaptive window	Not explicitly defined	avg(last 3 trips) + 2 min buffer, min 3 min, max 8 min